Exam Date	30/11/2025	Day	Sunday	Time	2:30-3:20
Student Name	[redacted]	Student ID No.	[redacted]	Section	11.30 - 10.30 د.عمر محمد بديل

GENERAL INSTRUCTIONS

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I have understood and abide with the exam policy. **Student's signature:**

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple Choice	CLO 1,8	20 (67%)			
2	Essay	CLO. 2-7	10 (33%)			
Total			30(100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

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Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	D	A	B	A	C	B	C	D	A	A

Question Number	Q11	Q12
Answer	C	D

22

Q1: If an event A satisfies $P(A) = 0.4$ then $P(\bar{A} \mid \Omega)$ equals
A) 0.3 B) 0.7 C) 0.4 D) 0.6 E) None

Q2: For some independent events A, B with $P(A) = P(B) = 0.1$ we have $P(A \cap B)$ equals
A) 0.01 B) 0.1 C) 0.2 D) 0.04 E) None

Q3: Let X be a random variable with zero mean. We have $E(3X^2 + X) =$
A) $2\,Var(X)$ B) $3\,Var(X)$ C) $9\,Var(X)$ D) $4\,Var(X)$ E) None

Q4: For every $x < 0$ we have $u(-x) + 3\delta(x) =$
A) 0 B) 2 C) $undefined$ D) 1 E) None

Q5 : Each ordered pair in $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$ has equal probability. The expected value of XY equals
A) 0.1 B) 0.2 C) 0.25 D) 0.3 E) None

Q6: Let X takes values 0 and 1 with $P(X = 0) = 0.9$ and let Y takes values 1 and 2 with $P(Y = 2) = 0.9$. If X and Y are independent, then $f(1,1) =$
A) 0.1 B) 0.01 C) 0.2 D) 0.02 E) None

Q7: For some continuous function f , if $\int_{-1}^{1} f(x)\,\delta(x)\,dx = 1$, then $f(0) =$
A) 0 B) -1 C) 1 D) -2 E) None

Q8: For the joint PDF $f(x,y) = x$ when $0 < x < \sqrt{2}$, $0 < y < 1$ and zero otherwise. We have $f_X(x) =$
A) 1 B) y C) $\sqrt{2}y$ D) x E) None

Q9: For a continuous random variable X with $P(1 < X < 2) = 0.3$, Then $P(0 < X < 2) - P(0 < X < 1) =$
A) 0.3 B) 0.4 C) 0.2 D) 0.1 E) None

Q10 : For two random variables X, Y with $Cov(X,Y) = 1$, $Cov(\pi - eX, e + \pi Y) =$
A) $-e\pi$ B) e C) $e\pi$ D) π E) none

Q11: Let X, Y be two independent random variables with $Var(X) = Var(Y)$ Then $Corr(X, Y) =$

A) 1 B) 1 C) 0 D) 3 E) None

Q12: A discrete random variable X with CDF $F(x) = a\,u(x+2) + a\,u(x-1)$

Then the constant a equals

A) 0 B) 0.25 C) 0.2 D) 0.5 E) None

X, y independed

$\forall$ $Cov(X,Y) = EXY - EXEy$

$= EXEy - EXEy$

$= 0$

$\exists a$

$\int_0^x du\,x+2 + \int_0^x a\,(u+1)$

$\int_0^x 2a = 1$

$(x-0)$

$2a = 1$

Q1:- Find $P(\bar{A}/\Omega)$

$P(A) = 0.4$

$P(\bar{A}) = 1 - 0.4 = 0.6$

$P(\Omega) = 1$

$$P(\bar{A}/\Omega) = \frac{P(\bar{A}) \times P(\Omega)}{P(\Omega)} = 0.6 \longrightarrow D$$

Q2:-

A, B Independent

$P(A) = P(B) = 0.1$

$P(A \cap B) = ??$

$P(A \cap B) = P(A) * P(B)$ (Independen)

$= 0.1 * 0.1 = 0.01 \longrightarrow A$

Q3:-

Zero mean

$\mu \rightarrow EX = 0$

Find

$E(3X^2 + X)$

Sol:-

$E(3X^2 + X) = 3EX^2 + EX = 3EX^2 = 3 \text{var} X \longrightarrow B$

$\text{var} X = EX^2 + (EX)^2$

$\text{var} X = EX^2 + (0)^2$

$\text{var} X = EX^2$

Q4:

$x < 0$, Find

$u(-x) + 3\delta(x) \longrightarrow$ Sol:- $u(-x)$, $x < 0$

$u(-x) = 1$

$3\delta(x) = 0$

$u(-x) + 3\delta(x) = 1 + 3(0) = 1 \longrightarrow D$

Q5:- $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$ has equal probability

$$E_{XY} = \sum_{R_X} \sum_{R_Y} xy\, f(x,y) = (0)(0) f(0,0) + (0)(1) f(0,1) + (1)(0) f(1,0) + (1)(1) * f(1,1)$$

$= 0 + 0 + 0 + (1)(1)(0.25)$

$= 0.25 \longrightarrow C$

$$\sum_{R_X} \sum_{R_Y} f(x,y) = 1$$

same

$f(0,0) + f(0,1) + f(1,0) + f(1,1) = 1$

$4f(1,1) = 1$

$f(1,1) = 1/4 = 0.25 \Rightarrow f(0,0) = f(0,1) = f(1,0)$

Q6:-

X, Y Independent

$X = \{0, 1\}$
$Y = \{1, 2\}$
$P(X=0) = 0.9$
$P(Y=2) = 0.9$

Find $f(1,1) \longrightarrow$ sol $F(1,1) = f_x(1) * f_y(1) \rightsquigarrow$ Independent

$P(X=1) = 1 - P(X=0) = 1 - 0.9 = 0.1$

$P(y=1) = 1 - P(y=2) = 1 - 0.9 = 0.1$

$F(1,1) = f_x(1) * f_y(1)$

$= 0.1 * 0.1 = 0.01 \longrightarrow B$

Q7:-

f continuous function

$\int_{-1}^{1} f(x)\,\delta(x) = 1$, Find $f(0)$

$\int_{-1}^{-1} f(x)\,\delta(x) = f(0)$

$0 \in [-1, 1]$

$f(0) = 1 \longrightarrow C$

Q8:- Joint pdf

$f(x,y) = x$, $0 < x < \sqrt{2}$, $0 < y < 1$

Find $f_x(x) = ??$

Sol:-

$f_{x(x)} = \int_{R_y} f(x,y)\,dy$

$= \int_0^1 x\,dy = x * [y]_0^1 = x\,(1-0) = \underline{x} \longrightarrow D$

or

$= x\,(1-0) = x$

Q9:- X continuous r.v

$P(1 < X < 2) = 0.3$

Find $= F(2) - F(1) = 0.3$

$P(0 < X < 2) - P(0 < x < 1)$

Sol:-

$P(a < X < b) = F(b) - F(a) \rightsquigarrow$ Continuous

$P(0 < X < 2) - P(0 < X < 1)$

$= F(2) - F(0) - [F(1) - F(0)]$

$= F(2) - F(0) - F(1) + F(0)$

$= F(2) - F(1) = 0.3 \longrightarrow A$

Q10:-

X, Y two r.v

Cov(X,Y) = 1

Find

Cov(π - eX, e + πY)

Cov(π - $\overset{a}{e}$X, e + $\overset{c}{\pi}$Y)

= (a)(c) Cov(X,Y)

= (-e)(π)(1) = -eπ ⟶ A

Q11:-

X, Y two Independent r.v

Var(X) = Var(Y)

Find

Corr(X,Y) = ??

Sol:-

X, Y ⟶ Independent

Cov(X,Y) = 0 , Corr(X,Y) = 0 ⟶ C

↳ $= \frac{Cov(X,Y)}{\sigma_x \sigma_y} = \frac{0}{\sigma_x \sigma_y} = 0$

Q12:-

X is a discrete r.v with CDF

F(X) = a(x+2) + a(x-1)

Find a ??

Sol:-

F(∞) = 1

a(x+2) + a(x-1) = 1

2a = 1

a = 1/2 = 0.5 ⟶ D

Q13:- Poisson distribution

Sol:-

⟶ $X \sim P(m)$, m = 3

$X \sim P(3)$

Find P(X=4)

$P(X=x) = \frac{e^{-m} m^x}{x!}$

$P(X=4) = \frac{e^{-4}(3)^4}{4!} = \frac{81e^{-4}}{24}$ ✓

Q14:- Binomial distribution

$X \sim B(n,p)$, $X \sim B(10, 0.8)$

n = 10

p = 80% = 0.8 , $q_{fail} = 1 - 0.8 = 0.2$

succes

Find P(X ≥ 1) = ??

P(X ≥ 1) = 1 - P(X < 1)

= 1 - P(X=0)

$= 1 - \binom{10}{0}(0.8)^0(0.2)^{10}$ ✓

$P(X=x) = \binom{n}{x} p^x q^{n-x}$

Exam Date	30/11/2025	Day	Sunday	Academic Year	2025/2026
Student Name	[redacted]	Student ID No.		Section	د. عمر بدير 10:30/11:30 خ ش ح

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Answer	B	D	A	B	C	D	A	B	D	C

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Q5 : Each ordered pair in $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$ has equal probability. The expected value of XY equals
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Q6: Let X takes values 0 and 1 with $P(X = 0) = 0.9$ and let Y takes values 1 and 2 with $P(Y = 2) = 0.8$. If X and Y are independent, then $f(1,1) =$
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Q7: For some continuous function f , if $\int_{-1}^{1} f(x)\,\delta(x)dx = 0$, then $f(0) =$
A) 0 B) 1 C) -1 D) -2 E) None

Q8: For the joint PDF $f(x,y) = y$ when $0 < x < 1$, $0 < y < \sqrt{2}$ and zero otherwise. We have $f_Y(y) =$
A) 1 B) y C) $\sqrt{2}y$ D) x E) None

Q9: For continuous random variable X with $P(1 < X < 2) = 0.1$, Then $P(0 < X < 2) - P(0 < X < 1)$ equals
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Q10 : For two random variables X, Y with $Cov(X,Y) = 1$, $Cov(\pi - eX, e - \pi Y) =$
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Q11: Let X, Y be two independent random variables with $Var(X) = Var(Y)$ Then $Corr(X, Y) =$

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Q12: A discrete random variable X with CDF

$$F(x) = a\,u(x+2) + 3a\,u(x-1)$$

Then the constant a equals

A) 0 B) 0.25 C) 0.2 D) 0.5 E) None



Q13: A call center receives an average of 3 calls per hour. Calculate the probability that exactly four calls arrive in a given hour!

$x=4$ $m=3$ ← Poisson

$x \sim P(m)$

$x \sim P(3)$

So we are $\frac{e^{-m} \cdot m^x}{x!}$

$$P(x) = \frac{e^{-3} \cdot (3)^4}{4!} = 0.168$$

Q14: A machine makes items, and each item has an 80% chance of being good. If we check 10 (n) items, what is the probability that at least one item is good?

$x \sim B(n, P)$ $x \sim B(10, \frac{4}{5})$ $P = 80\% = \frac{4}{5}$

$q = 1 - P = \frac{1}{5}$

$P(x) = \binom{n}{x} P^x q^{n-x}$

$= P(x \geq 1)$

$1 - P(x \leq 1)$

$1 - P(x=0)$

$1 - \binom{n}{x} P^x q^{n-x}$

$1 - \left(\binom{10}{0} P^0 q^{10-0}\right)$

$1 - \left(1 \left(\frac{4}{5}\right)^0 \left(\frac{1}{5}\right)^{10}\right) = 0.999 = 1 - 0.2^{10}$

$P(x \geq 1) = 0.999$

النموذج (2)

Q1:-

Find $P(\bar{A}|\Omega)$

$P(A) = 0.3$

$P(\bar{A}) = 1 - 0.3 = 0.7$ (P(A))

$P(\Omega) = 1$

Soli-

$$P(\bar{A}|\Omega) = \frac{P(\bar{A}) * P(\Omega)}{P(\Omega)} = \underline{0.7} \longrightarrow B$$

Q2:-

A, B Independent

$P(A) = P(B) = 0.2$

$P(A \cap B) = ??$

Soli-

$P(A \cap B) = P(A) * P(B)$

$= 0.2 * 0.2$

$= 0.04 \longrightarrow D$

Q3:-

mean

$E[X] = \mu = 0$

Find

$E(2x^2 + x)$

Soli-

$Ex = 0$, $E(2x^2 + x) = 2Ex^2 + \cancel{Ex}^{\to 0}$

$Var X = Ex^2 + \cancel{(Ex)^2}^{\to 0}$

$\therefore 2 Var X \longrightarrow A$

Q4:-

$x \geq 0$ Find $2u(-x) + 3\delta(x)$

Soli-

$2u(-1) + 3\delta(0) = 2(1) + (0)(3) = 2 \longrightarrow B$

Q5:- [redacted]

$\sum\sum f(x,y) = 1$

نفس حل النموذج السابق

same Prob.

$f(0,0) + f(1,0) + f(0,1) + f(1,1) = 1$

$4\, f(1,1) = 1/4 = f(0,1) = f(1,0) = f(0,0)$

Ex. $\sum\sum xy\, f(x,y) = 0.25 \longrightarrow C$

Q6:-

$X: \{0,1\}$
$Y: \{1,2\}$
$P(X=0) = 0.9$
$P(X=2) = 0.8$
X, Y independent

$f(1,1) = ??$

$f_{(1,1)} = f_x(1) * f_y(1)$ ⇝ independent

$P(X=1) = 1 - P(X=0) = 0.1$

$P(X=1) = 1 - P(X=2) = 0.2$

$= 0.1 * 0.2$

$0.02 \longrightarrow D$

Q7: f is continuous function

$$\int_{-1}^{1} f(x)\,\delta(x) = 0$$ Find $f(0)$

Solv

$x_0 = 0 \in [-1,1]$ ✓

$$f(0) = \int_{-1}^{1} f(x)\,\delta(x) = 0 \longrightarrow A$$

Q8 Pdf Joint $f(x,y) = y$, $0<x<1$, $0<y<\sqrt{2}$

Find $f_y(y)$

Solv

$$f_y(y) = \int_{Rx} f(x,y)\,dx = \int_0^1 y\,dx = y(1-0) = y \longrightarrow B$$

Q9:-

X continuous r.v $P(1<X<2) = 0.1$

Find $P(0<X<2) - P(0<X<1) = ?$

Solv

$P(a<x<b) = F(b) - F(a)$

$P(0<X<2) - P(0<X<1) = F(2) - F(0) - [F(1) - F(0)]$

$= F(2) - F(0) - F(1) + F(0) = F(2) - F(1) = 0.1 \longrightarrow D$

Q10:-

$Cov(x,y) = 1$

Find $Cov(\pi - ex, e - \pi y)$

Sol:-

$Cov(\pi - ex, e - \pi y) = (-e)(-\pi)\, Cov(x,y) = e\pi \longrightarrow C$

Q11:- X, Y Independent ∩, Find Corr(nay)

$Corr(x,y) = 0 \longrightarrow A$

Q12:-

$F(x) = a\,u(x+2) + 3a(x-1)$, Find a

Sol:-

$F(\infty) = 1$,

$a\,u(x+2) + 3a(x-1) = 1$

$a + 3a = 1$

$4a = 1$

$a = 1/4 = 0.25 \longrightarrow B$

Q13:- Poisson distribution

$x \sim P(m)$, $m = 3$

$x \sim P(3)$

$P(X = x) = \dfrac{e^{-m} * m^x}{x!}$

$P(X = 4) = \dfrac{e^{-3} * 4^3}{4!}$

$= \dfrac{81\, e^{-4}}{24}$

Q14:- Binomial distribution

$x \sim B(n,p)$, $n = 10$

$x \sim B(10, 0.8)$, $p = 80\% = 0.8$

$P(x \geq 1) = 1 - P(X = 0)$, $q = 1 - 0.8 = 0.2$

$= 1 - \binom{10}{0}(0.8)^0(0.2)^{10}$